

Modeling Human Activity States Using Hidden Markov Models

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1. Background and Motivation

Human Activity Recognition (HAR) is a vast research area that has many applications in the smart home, fitness and health monitoring sectors. This project builds an automatic recognition system for four activities (walking, standing, still and jumping) using only sensor data from the smartphone. This is significant in resource-poor settings where specialized wearables are not available. Human Activities are activities that happen over time, and the actual activity state can only be deduced from historical measurements of the sensors, which is solved by a Hidden Markov Model (HMM).

2. Data Collection and Preprocessing

Data was collected using Sensor Logger app on a phone running at 100 Hz. There were four activities (8-10s per clip) for a total of 52 clips. No harmonization of the sampling rate was required as one device was used. These were exported as two files, Accelerometer.csv and Gyroscope.csv, that were made into a 6 column array (3 on each axis, where one axis is accel, and the other is gyro).

Activity	Clips	Duration	Phone Position
Walking	13	8-10 sec	Hand at waist level
Standing	13	8-10 sec	Hand at waist level
Still	13	8-10 sec	Flat on surface
Jumping	13	8-10 sec	Hand at waist level

Table 1: Data collection summary.

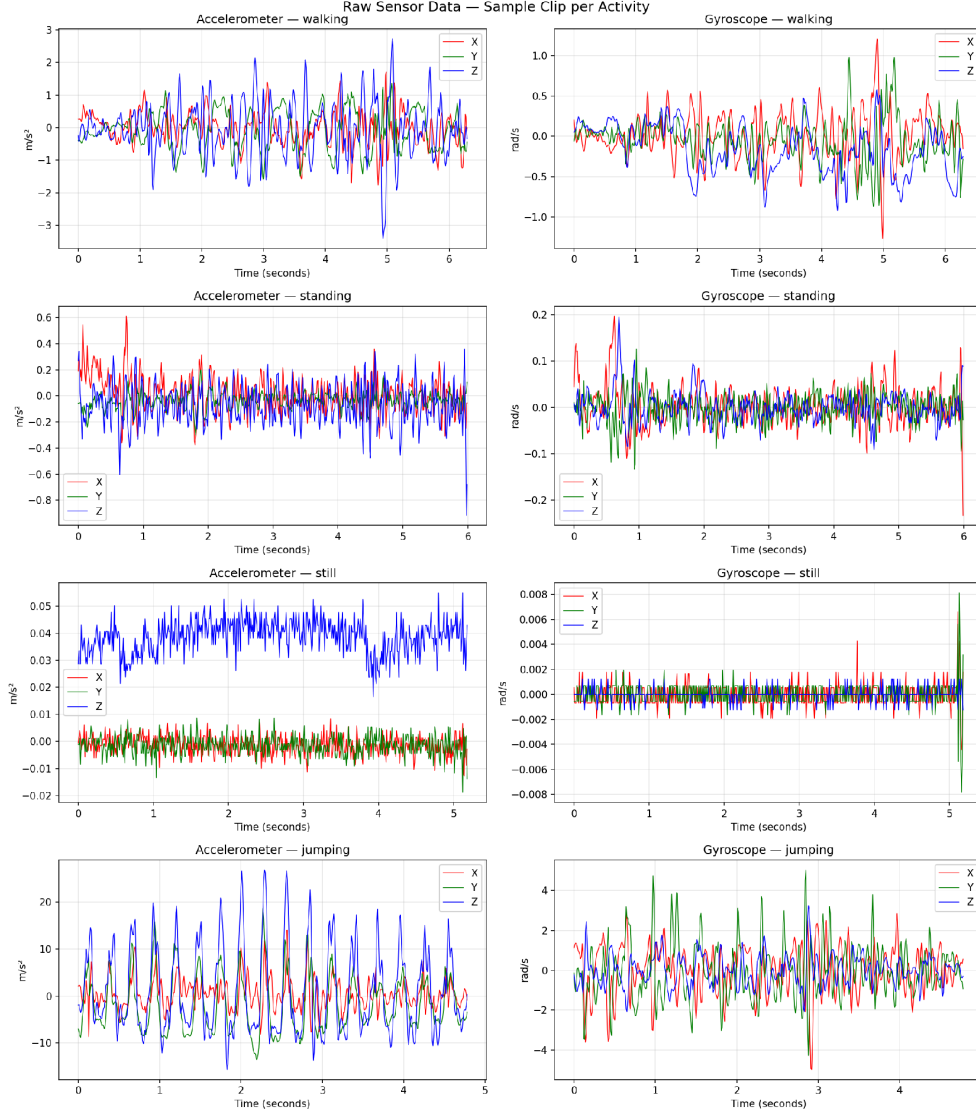


Figure 1: Raw accelerometer and gyroscope signals per activity.

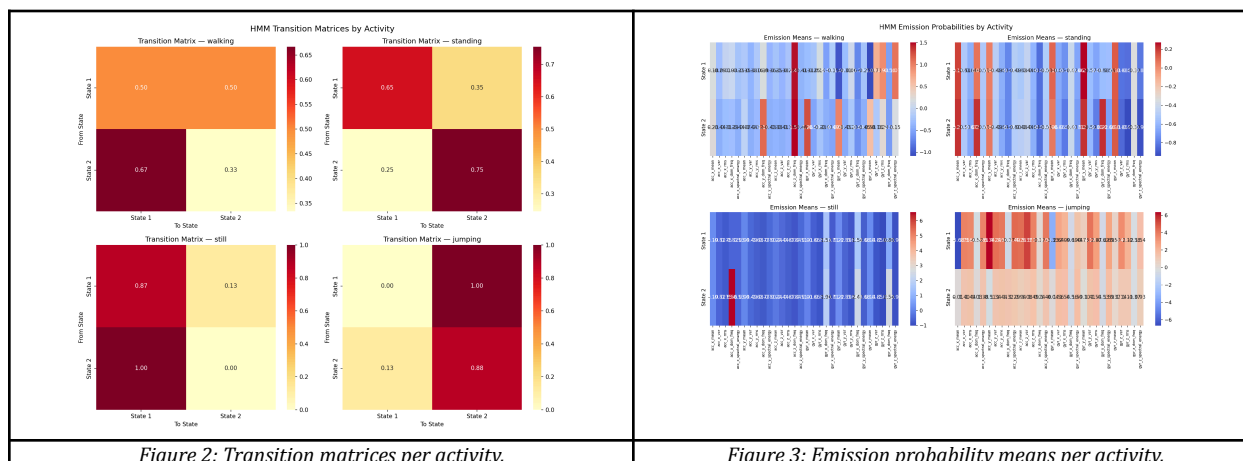
3. Feature Extraction

30 features were extracted per clip (5 features per axis and 6 axes). In the time domain, it is the mean (average level), the variance (fluctuation of the signal) and the RMS (total energy). The frequency domain: dominant frequency (main frequency movement rhythm) and spectral energy (total frequency power). These features are different intensities and pattern/rhythm of movement and are different between activities. To compensate for the differences in scale between the features in the accelerometer (m/s^2) and the gyroscope (rad/s), all features were normalized using Z-score standardization (StandardScaler).

4. HMM Setup and Implementation

A single Gaussian HMM was learnt for each activity with the use of hmmlearn. There are 2 hidden states for each model indicating sub-phases of the activity. It was trained with Baum-Welch algorithm and $\text{tol} = 1\text{e-}4$, these parameters stopped learning when the log-likelihood improvement was below 0.0001, which is more meaningful than a fixed number of iterations. The data were stratified sampled 2 parts: 10:1 training &

testing. For decoding, the test clips were compared to all four models, with the one model with the best log-likelihood score being chosen as the model to be selected as a prediction.

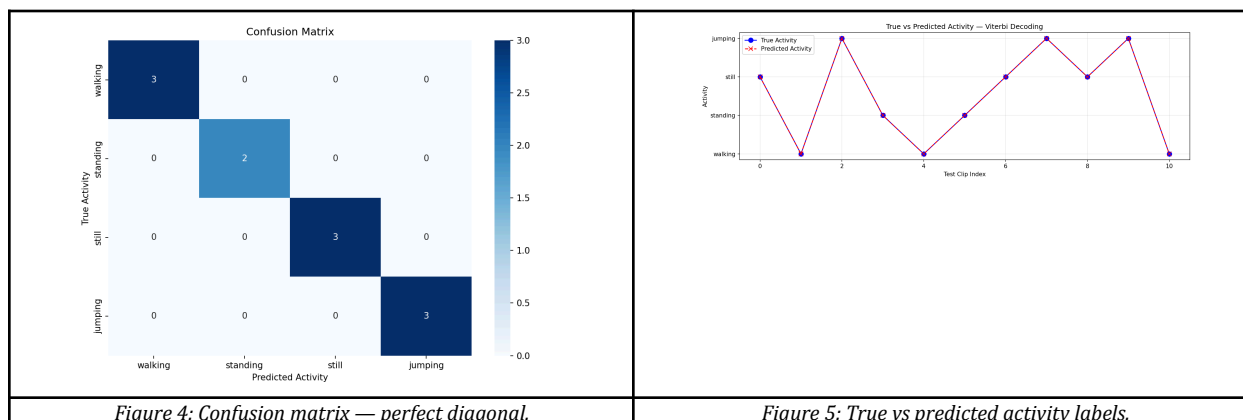


5. Results and Interpretation

Activity	Samples	Sensitivity	Specificity	Accuracy
Walking	3	100.0%	100.0%	100.0%
Standing	2	100.0%	100.0%	100.0%
Still	3	100.0%	100.0%	100.0%
Jumping	3	100.0%	100.0%	100.0%
Overall	11	100.0%	100.0%	100.0%

Table 2: Performance metrics on test set.

Also, the model was capable of correctly classifying two unseen clips recorded in a different session: teste_walking_01, Walking, and teste_standing_01, Standing, which showed that the model was able to generalise to other data.



The transition matrices (Figure 2) show that the self-transition is high for both still and walking, and balanced transition for walking due to the rhythmic nature of walking. The difference in activities is reflected in the emission probabilities (Fig. 3), and a good indication that the four classes are well demarcated.

6. Discussion and Conclusion

Both the test set and unseen recordings were 100% accurate. Near-zero vs. large spikes in the signal allowed easy distinction between still and jumping. The rhythmic oscillations were observed during walking and not during standing and they differentiated walking and standing. The data was divided into the 80/20 and there are not many training samples, so there is limited convergence of two models. There are some other enhancements to be made, including more signal magnitude area (SMA) and more clips gathered as well as testing on other participants. It has been demonstrated that Gaussian HMMs learned by Baum-Welch algorithm and decoded by the Viterbi algorithm are well suited for sequential human activity recognition with a specific sensor set consisting of smartphone sensors.

Task Allocation

All tasks completed individually by Josephine Duba Kanu (class decision to work individually).

Task	Completed by
Data collection	Josephine Duba Kanu
Preprocessing	Josephine Duba Kanu
Feature extraction	Josephine Duba Kanu
HMM implementation	Josephine Duba Kanu
Evaluation and metrics	Josephine Duba Kanu
Report writing	Josephine Duba Kanu
GitHub repository	Josephine Duba Kanu